

ASSIGNMENT 4:

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CLASS : SY BTECH IT

INHERITANCE AND POLYMORPHISM

/* 4.A) Method Overloading:

Implement program for a geometric shapes

application that utilizes function overloading for calculating properties such as area and

perimeter. Implement overloaded methods for computing area and perimeter for different

shapes (e.g., rectangles, circles, triangles) with varying sets of parameters.

*/

```
import java.util.*;
```

```
class Shape {
```

```
    //Method to calculate Area of circle
```

```
    void area(double r) {
```

```
        double a= 3.14*r*r;
```

```
        System.out.println("Area Of Circle is "+a);
```

```
    }
```

```
    //Method to calculate circumference of circle
```

```
    void perimeter(double r) {
```

```
        double per= 2*3.14*r;
```

```
        System.out.println("Circumference of Circle is "+per);
```

```
    }
```

```
    //Method to calculate Area of Rectangle
```

```
    void area(double l,double b) {
```

```
        double a= l*b;
```

```
        System.out.println("Area Of Rectangle is "+a);
```

```
    }
```

```
    //Method to calculate Perimeter of Rectangle
```

```
    void perimeter(double l,double b) {
```

```
        double per= 2*(l+b);
```

```
        System.out.println("Perimeter of Rectangle is "+per);
```

```
    }
```

```
    //Method to calculate Area of Triangle
```

```
    void area(double a,double b,double c) {
```

```

        double s=(a+b+c)/2;
        double ar= Math.sqrt(s*(s-a)*(s-b)*(s-c));
        System.out.println("Area Of Circle is "+ar);
    }
    //Method to calculate Perimeter of Triangle
    void perimeter(double a,double b,double c) {
        double per= a+b+c;
        System.out.println("Perimeter of Triangle is "+per);
    }
}

```

```

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int ch;
        do {
            System.out.println("\n**** MENU ****");
            System.out.println("1. Circle");
            System.out.println("2. Rectangle");
            System.out.println("3. Triangle");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    Shape circle=new Shape();
                    System.out.println("Enter the radius of circle:");
                    double r=sc.nextDouble();
                    circle.area(r);
                    circle.perimeter(r);
                    break;
                case 2:
                    Shape rec=new Shape();
                    System.out.println("Enter the length and breadth of Rectangle:");
                    double l=sc.nextDouble();
                    double b=sc.nextDouble();
                    rec.area(l,b);
                    rec.perimeter(l,b);
                    break;
                case 3:

```

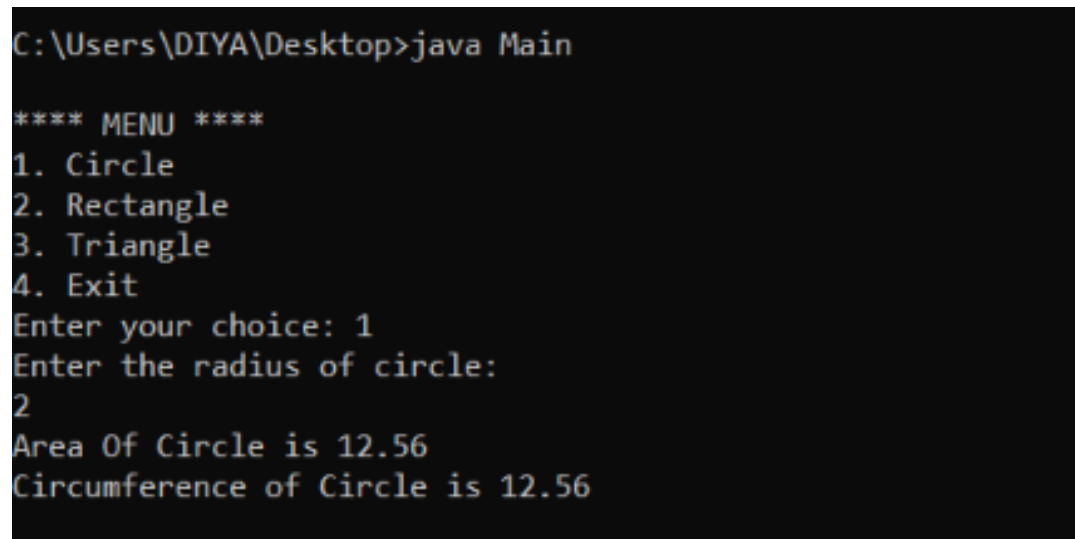
```

        Shape tri=new Shape();
        System.out.println("Enter the three sides of triangle:");
        double a=sc.nextDouble();
        double b1=sc.nextDouble();
        double c=sc.nextDouble();
        tri.area(a,b1,c);
        tri.perimeter(a,b1,c);
        break;
    case 4:
        System.out.println("Successfully Exited");
        break;
    default:
        System.out.println("Invalid Choice");
    }

    } while (ch != 4);
}

```

OUTPUT:



```

C:\Users\DIYA\Desktop>java Main

**** MENU ****
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 1
Enter the radius of circle:
2
Area Of Circle is 12.56
Circumference of Circle is 12.56

```

```
**** MENU ****
```

1. Circle
2. Rectangle
3. Triangle
4. Exit

Enter your choice: 2

Enter the length and breadth of Rectangle:

2 3

Area Of Rectangle is 6.0

Perimeter of Rectangle is 10.0

```
**** MENU ****
```

1. Circle
2. Rectangle
3. Triangle
4. Exit

Enter your choice: 3

Enter the three sides of triangle:

3 4 5

Area Of Circle is 6.0

Perimeter of Triangle is 12.0

```
**** MENU ****
```

1. Circle
2. Rectangle
3. Triangle
4. Exit

Enter your choice: 4

Successfully Exited

C:\Users\DIYA\Desktop>_